PAPER_228: Westerlund 2 Super Star Cluster â€” MUGE with High-Density OB-Supergiant Wind Feedback (10Ã— LMC)

Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grok_share_8d951e12.txt extraction â€” Doc 6) **Date:** March 2026 **Classification:** Novel MUGE â€” Extreme OB-Supergiant Wind Density Comparison **Status:** Proof-Quality Whitepaper

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Abstract

Westerlund 2, the most massive super star cluster in the Milky Way (Carina-Sagittarius Arm, ~10 kly), is modelled with a 9-term MUGE. The distinguishing novelty is the OB-supergiant stellar wind density $\rho_{wind} = 10^{-20}$ kg/mÂ^3 â€” ten times denser than the LMC Tapestry (PAPER_227). This quantitative distinction within the shared $a_{wind} = \rho_{wind} v_{wind}^2 / \rho_{fluid}$ framework makes Westerlund 2 the highest-wind-density entry in the MUGE stellar-wind family.

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1. Physical System

Westerlund 2 (Wd2) contains ~300 massive O/B stars in a ~ 4 ly core:

Parameter	Value
Distance	~ 10 kly (Milky Way, Carina arm)
M_{init}	$30{,}000 M_\odot$
M_{gas}	$100{,}000 M_\odot$
M_{dot_factor}	$M_{gas}/M_{init} = 3.33$
r	10 ly
B	10 μT
τ_{SF}	2 Myr (rapid OB-dominated SF)
ρ_{wind}	10^{-20} kg/mÂ^3
v_{wind}	2000 km/s

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2. Key Distinction: Wind Density Factor

Comparative Table

System	M_{init}	ρ_{wind}	τ_{SF}	a_{wind} (m/sÂ^2)
Tapestry LMC (PAPER_227)	$240 M_\odot$	10^{-21} kg/mÂ^3	5 Myr	4×10^3
Westerlund 2 (PAPER_228)	$30{,}000 M_\odot$	10^{-20} kg/mÂ^3	2 Myr	4×10^4
Ratio	$125\times$	$10\times$	$0.4\times$	$10\times$

The 10Ã— higher wind density in Westerlund 2 directly scales the wind acceleration, making it the most wind-dominated system in this MUGE subset.

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3. Gas-Accreting Mass (shared with PAPER_227)

$$M(t) = M_{init}\left(1 + \frac{M_{gas}}{M_{init}} e^{-t/\tau_{SF}}\right)$$

$$\dot{M}_{\rm UQFF} = \dot{M}_0\big(1 - [SSq]\, e^{-\kappa\Delta t}\big)\,,\quad \kappa = 5.0\times10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

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Nameg_\text{UQFF}(r) = g_\text{MUGE}(r)\cdot\Bigl(1 - [SSq]\cdot U_{b_i},/,F_U(r,t)\Bigr), \quad [SSq] = 0.57Name

With $M_{dot\_factor} = 3.33$ (vs 41.7 for Tapestry): smaller fractional gas reservoir relative to the massive cluster.

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4. Physical Rationale for Dense Wind

Westerlund 2 hosts ~ 300 O- and B-type supergiants, including WR 20a (a $\sim 83M_\odot + 82M_\odot$ WN binary). These stars produce winds with mass-loss rates $\dot{M}_{star} \approx 10^{-5}$ to $10^{-4}M_\odot$/yr. The collective wind deposits $\sim 10^{-20}$ kg/mÂ^3 in the 10 ly shell around the cluster core.

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5. Calculator Class

```
class Westerlund2MUGEStellarWindCalculator(_CP3Calculator):
    """PAPER_228: Westerlund 2 â€” 9-term MUGE, rho_wind=1e-20 (10x LMC), OB-supergiant
    # Session 58 â€” grok_share_8d951e12.txt Doc 6
```

Located in `CondensedPhysics3.py` (Session 58).

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6. Conclusion

Westerlund 2 establishes the upper bound of the MUGE stellar wind family with $\rho_{wind} = 10^{-20}$ kg/mÂ^3, placing it 10Ã— above the LMC Tapestry. The shared a_{wind} formula with different physical parameters demonstrates the parametric flexibility of the MUGE wind extension while remaining anchored to observationally motivated values.

Source: grok_share_8d951e12.txt â€” Doc 6 (Westerlund 2 OB Wind MUGE)

UQFF computed: MUGE buoyancy ratio U_bi/F_U = [SSq]×?×r^2/GM = 5.7e-1×5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s^2 at r_ISCO.